# Rajiv Choudhury

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## Summary

Incoming CS graduate student at **UC Davis** (Fall 2026) and recent computer science graduate from IIIT Guwahati with experience in on-device AI, embedded systems, and energy-efficient computing. Interested in the intersection of **parallel computing** and **IoT/embedded devices** to enhance **ML inference efficiency** on constrained hardware such as smartwatches, TVs, and digital appliances. Experienced in developing optimized AI systems and conducting research with **IISc Bengaluru** and **IBM Research**. Additionally, conducted research on embedded AI for the Android platform with **Samsung**.

## Education

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|---|---|---|
| Graduate Student | MS (Computer Science) at **UC Davis** | |
| Class of 2024 | B.Tech (Computer Science Engineering) at **IIIT Guwahati** | (CPI: 9.13/10) |
| | *Relevant Coursework:* Operating Systems, Computer Organisation, Machine Learning, Deep Learning, Computer Networks, Theory of Computaion, Data Structure & Algorithm | |
| Class of 2020 | Class 12th MBS Public School (Affilitated with CBSE) | (Grade: 90.80%) |
| Class of 2018 | Class 10th DAV Public School (Affilitated with CBSE) | (Grade: 90.60%) |

## Work Experience

**Software Engineer — Samsung R&D Institute India - Delhi** Jul 2024 – Aug 2026

- Developed and maintained **Tizen OS** application features using **C++** and **C#**.
- Developed a **USB preset** feature for **E-paper signage devices**, enabling zero-contact device setup through configuration files.
- Implemented an early-warning feature for certificate expiration across **10,000+ enterprise displays**, which received the team's **Best Feature Award**.
- Maintained and developed multiple modules for **Tizen OS**, including the **Power Saving Module**, achieving up to **80% power savings** across enterprise displays.
- Developed and maintained a **logger service** for collecting logs from Android devices, enabling efficient post-sale diagnostics and reducing customer support effort.
- Developed an on-device AI feature integrating an offline **Speech-to-Text model (OpenAI Whisper)** on Android Interactive Displays for Samsung's AI Assistant project.
- Built Android applications for Interactive Displays using **Jetpack Compose**, improving user experience and enhancing application performance by **25%**.
- Developed an Android application for hotel self check-in using **Aadhaar-based e-verification** and face verification for seamless guest onboarding.

**Research Intern — Centre for Networked Intelligence, IISc Bengaluru** Jan 2024 – Jun 2024

- Studied **NVIDIA GPU architecture** and parallel computing concepts for high-performance computing workloads.
- Conducted experiments that outperformed Linux CPU governors in energy efficiency by up to **80%** using the proposed scheduling method.
- Collaborated with **IBM Research** on GPU scheduling techniques and matrix multiplication optimization for parallel architectures.
- Implemented Logistic Regression and Neural Network models in **CUDA C++** for training and inference, and compared their performance with the **PyTorch** framework.

**Research Intern — IIIT Guwahati** May 2022 – Jul 2022

– Worked on **Brain-Computer Interface (BCI)** systems for EEG signal analysis and classification.

– Studied and implemented **MLP classifiers** and Neural Networks for brain signal processing tasks.

## Projects

**Classification Simulation (Brain Signal Analysis)** Tech: Python, Tkinter

– Engineered a TRL-4 prototype to classify EEG brain signals using `MLPClassifier`, enabling patients with motor neuron disease to communicate with their environment.

– Utilized `EMOTIV INSIGHT` hardware for real-time data collection, classifying test data with **90% accuracy**.

**Fashion Recommender** Tech: Python, Streamlit, TensorFlow

– Created an application to recommend fashion products using a pre-trained `ResNet` model trained on over 1 million images.

– Implemented a K-Nearest Neighbors algorithm for image similarity, achieving **98% prediction accuracy**.

**On-Device Whisper AI Assistant** Tech: Kotlin, Jetpack Compose, C++, Whisper

– Developed an on-device AI feature integrating OpenAI's Whisper model for offline speech-to-text transcription on Android Interactive Displays.

– Optimized inference pipeline for low-latency performance, contributing to Samsung's AI Assistant project.

## Publications

Choudhury, Rajiv et al. (Apr. 2024). "A Brain-Computer Interface Controlled Assistive Device for Persons with Motor Neuron Disease". In: *2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)* 99.18, pp. 2200–2300. URL: https://ieeexplore.ieee.org/abstract/document/10621489/.

Priya, Aditya et al. (June 2024). "Energy-minimizing workload splitting and frequency selection for guaranteed performance over heterogeneous cores". In: *Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems* 99.18, pp. 2200–2300. URL: https://dl.acm.org/doi/pdf/10.1145/3632775.3661968.

## Skills

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| **Programming Languages** | Python, C++, Java, CUDA C++, C# |
| **Deep Learning Frameworks** | TensorFlow, PyTorch |
| **Libraries & Tools** | NumPy, Pandas, Scikit-learn, Git, Docker |
| **Platforms / Operating Systems** | Tizen, Linux, Windows, Android, nvidia-smi |